TYREMIND

Causal tyre intelligence — isolating true wear from everything that moves a lap time

Every published tyre model is graded on how well it predicts lap times. **None of them checks whether the reason it gives is the right one.** On 3 of the 4 real 2024 races analysed, the standard method reports that tyres get faster the longer they run.

95.5%
Error reduction recovering a hidden rate
25 synthetic sessions, 75 comparisons

0.22 ms
Per-lap live update
flat in session length, 921 laps

7/8
Circuits whose rotation the physics recovered
never told the answer

22.7
RUL RMSE on NASA turbofans, same estimator
all 100 test engines, no tyre-specific code

THE PROBLEM

The standard method says F1 tyres get faster as they wear

Fit a line through lap time against tyre age — what every team, every broadcaster and every public analysis does. On the 2024 Italian Grand Prix it returns a negative degradation rate on both compounds.

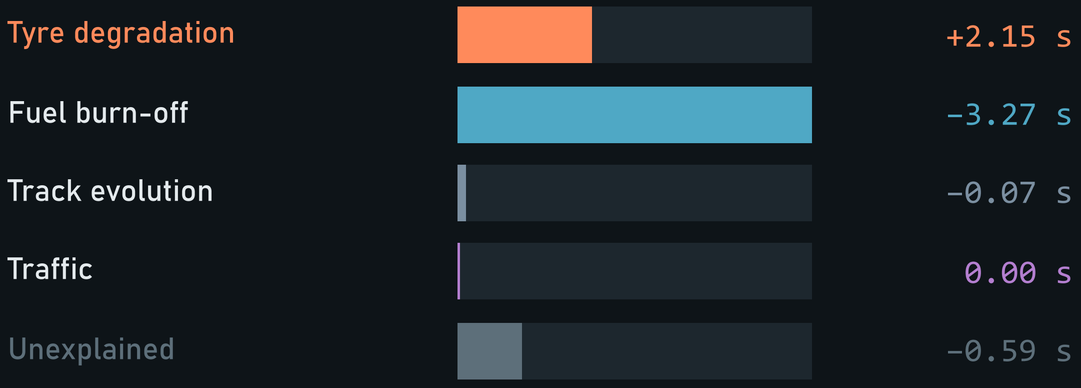
-0.004	-0.034	0.074	0.113
Naive, HARD	Naive, MEDIUM	TyreMind, HARD	TyreMind, MEDIUM
s/lap — tyre improving with use	s/lap — also impossible	±0.018 s/lap, same laps	±0.023 s/lap, same laps

> This is not a subtle miscalibration a practitioner can mentally correct. It is a claim that **tyres improve with use**, produced from real Grand Prix data by the method the field actually uses. On a race stint the confounders are not noise around the signal — **they are larger than the signal, and they point the other way.**

live API · /api/session/2024-monza-R · 921 laps, 20 cars

WHY IT HAPPENS

Four things move a lap time. Only one of them is the tyre.



What the stopwatch saw **−1.78 s**

One real stint, decomposed

Ricciardo, Monza 2024 — 41 laps on the hard tyre, laps 13 to 53. The car was **1.78 seconds faster**, so a stopwatch says the tyre is fine.

It is not. The hard tyre **lost 2.15 seconds** across that stint. The gain came from elsewhere — mostly fuel burn-off, worth 3.27 seconds.

The car got 2.7 kg lighter each lap, and each kilogram is worth about 0.030 s. That is **0.081 s/lap of free pace**, larger than most degradation rates and pointing the opposite way.

Reading pace alone would miss a degrading tyre completely. This is the screen the product opens on.

/api/session/2024-monza-R/decompose-run · driver RIC, 41-lap stint on HARD

THE GAP IN THE FIELD

Every published model is graded on lap-time error. That metric is blind to this failure.

> Fuel burn-off, track evolution and tyre degradation all move lap time smoothly with lap number. Many wrong decompositions sum to the same right total. A model can predict lap times almost perfectly while blaming entirely the wrong cause — and no lap-time metric will notice, because the total is right.

PRIOR WORK	WHAT IT DOES	WHAT IT LEAVES OPEN
Cappello & Hoegh arXiv:2512.00640, Nov 2025	Bayesian state-space tyre degradation. Fuel + latent tyre pace, pit stops as resets, FastF1.	**Single driver, single race.** No traffic, no track evolution, no physics. Reports compounds as statistically indistinguishable.
Pitwall arXiv:2607.06495, 2026	Calibrated Monte Carlo race simulator + gated LLM briefings. Ran live at two 2026 Grands Prix.	Own stated gaps: **no car telemetry**, approximate fuel, thermals in the noise. Predicts outcomes, not causes.
Explainable tyre energy arXiv:2501.04067	XGBoost + deep models on tyre energy, with SHAP and counterfactuals.	Uses **private Mercedes-AMG telemetry.** Not reproducible from public data.
Heilmeier et al. TUMFTM/race-simulation	The open-source reference race simulator and pit-stop optimiser.	Takes a degradation rate as **given** — which is precisely the number we produce.

The gap is not carelessness. Testing attribution needs a ground truth motorsport cannot supply: public F1 data has no tread depth, no rubber mass, no wear sensor. So the field scores what it can measure.

THE ACTUAL HARD PART

Three collinearities. Only the first is in the brief.

01 · RESOLVED BY ASSUMPTION

Fuel vs degradation

Both linear in laps completed. Within one run they are **not separable at all** — not poorly separable.

Pinned by a physical prior, 0.030 s/kg × 2.7 kg/lap, carried as state so its uncertainty **widens every published interval**.

02 · RESOLVED BY ASSUMPTION

Track evolution vs a uniform rate shift

Not anticipated. Found by chasing a stubborn −0.013 s/lap bias that survived removing the cliff, scrubbed sets and traffic.

Resolved with a saturating basis $A(1-e^{-kL})$ and an informative amplitude prior. Bias → +0.0012.

03 · RESOLVED BY EVIDENCE

Tyre age vs session lap

They advance together. One car cannot tell an ageing tyre from a changing session.

Fit the whole field. Cars pit on different laps, so pit stagger is a natural experiment — free, no prior. A single-driver model cannot use it.

```
degradation moves by  +c·(a₀ + L − L₀)
track       moves by  −c·L
──────────────────────────────────
net                    c·(a₀ − L₀)

← constant WITHIN a run, which is
  exactly what the run intercept absorbs
```

Why the second one is invisible until you write it down

Shift every degradation rate by c and the track slope by $-c$. The difference is constant within a run, so the run intercept swallows it and the two are **structurally indistinguishable**. Scrubbed sets do not help.

Two of three are resolved by assumption, not by data. We say so, and we measure what the assumptions cost — the next slide is that measurement.

WHAT THE ASSUMPTIONS COST

We re-fit everything under wrong priors and publish what happens

4 synthetic sessions × 3 compounds = 12 comparisons per row. Grouped by what is perturbed, not sorted by score.

VARIANT	RATE MAE	BIAS	MEAN SD	COVERAGE
baseline	0.0047	+0.0043	0.017	100%
live at 75% of the session	0.0049	+0.0035	0.018	100%
half the field	0.0073	+0.0056	0.018	100%
wide priors	0.0055	+0.0053	0.032	100%
track prior −1sd	0.0023	+0.0000	0.017	100%
track prior +1sd	0.0091	+0.0091	0.017	100%
fuel prior −1sd	0.0113	-0.0113	0.017	100%
fuel prior +1sd — worst case	**0.0199**	+0.0199	0.017	100%

Four readings

- **The worst case is still 4.9× better than naive** — 0.0199 against 0.0966. The assumptions carry the answer, but not alone.
- **The track prior traces a clean slope** — bias +0.0000, +0.0043, +0.0091 across −1sd, baseline, +1sd: about 0.0045 s/lap per prior sd, monotone. **−1sd scoring best is cancellation, not superiority** — the prior sits exactly on the generator's true 0.90 s, and moving it down offsets a small positive bias.
- **Wide priors widen the posterior sd by 1.9×** — 0.017 → 0.032. Being less sure produces wider intervals rather than a different answer.
- **Coverage holds at 100% throughout.** Even when the prior is wrong, the interval still contains the truth.

Why 0.0047 here and 0.0044 on the headline slide? Different samples, not different answers: the headline benchmark runs **25 sessions**, while re-fitting eight variants is costly enough that this table runs **4**. Every perturbation is read against this table's own baseline.

> **A team with real fuel telemetry should replace our prior with their measurement.** The architecture supports it directly, and doing so removes the single largest assumption in the method. We say this in the product, not only in the appendix.

experiments/results/exp02_prior_sensitivity.json

A hierarchical state-space model in tyre age, fitted across the whole grid

```
y[d,t] = alpha[r]             run intercept: car, setup, start fuel
       - A · g(t)             track evolution, shared field-wide
       + theta[t]             residual track wobble
       + s[d,t]               latent tyre loss  ← the deliverable
       - phi · L[d,t]         fuel burn-off
       + gamma · TI[d,t]      traffic index
       + eps                  heavy-tailed observation noise

latent tyre evolves in TYRE AGE, not session lap:
s[d,a+da]    = s[d,a] + da·rate[d,a] + noise
rate[d,a+da] = rate[d,a] + noise

and stints pool towards their compound:
rate[stint] ~ N(compound_baseline[c], sd²)
```

A cliff needs no special machinery

Because the rate is itself a free state, degradation that accelerates simply appears as `rate` rising. The same structure represents a plateau or a recovery. **Nothing assumes a cliff exists** — which matters, because a model with a hard-coded cliff term will find one whether or not it is there.

Coefficients are states with zero process noise

For a constant, that *is* the Bayesian posterior. Physical priors enter as `P0` and the filter propagates their uncertainty into the tyre posterior automatically — no bootstrap, no second pass. It is why the previous slide's sd column responds to prior width.

src/tyremind/models/ssm/tyre_ssm.py · docs/research/05_STATISTICAL_ARCHITECTURE.md

THE ARCHITECTURAL CHOICE

A Kalman filter, not a sampler — so the same model runs live

01

An exact likelihood

The model is linear-Gaussian given six variance hyperparameters, so the marginal likelihood is closed-form through the prediction error decomposition. L-BFGS-B optimises it directly. No convergence diagnostic stands between the model and a number.

02

Analytic posteriors

For every state, including the coefficients. No sampling error. An interval reported at 0.018 s/lap is that width because the algebra says so, not because a chain happened to explore that far.

03

An incremental mode

The forward pass alone *is* the real-time estimator. A sampler has no such mode, and re-running one every lap is not an option on a pit wall.

The novelty is architectural, and we say so

Kalman filtering is decades old, and live race analytics already exist. What is worth defending is that **the same estimator serves both modes**. The retrospective curve an engineer studies on Friday evening and the live number on the pit wall on Sunday come from one model — not from a fast approximation of a slow one. Filtered and smoothed estimates are kept strictly separate, and the live estimator agrees with the batch fit to **0.0006 s/lap** at worst — two independent implementations of the same model, not the same fit read twice.

0.22 ms

Mean per-lap update

0.74 ms worst case, 921 laps

5.8 s

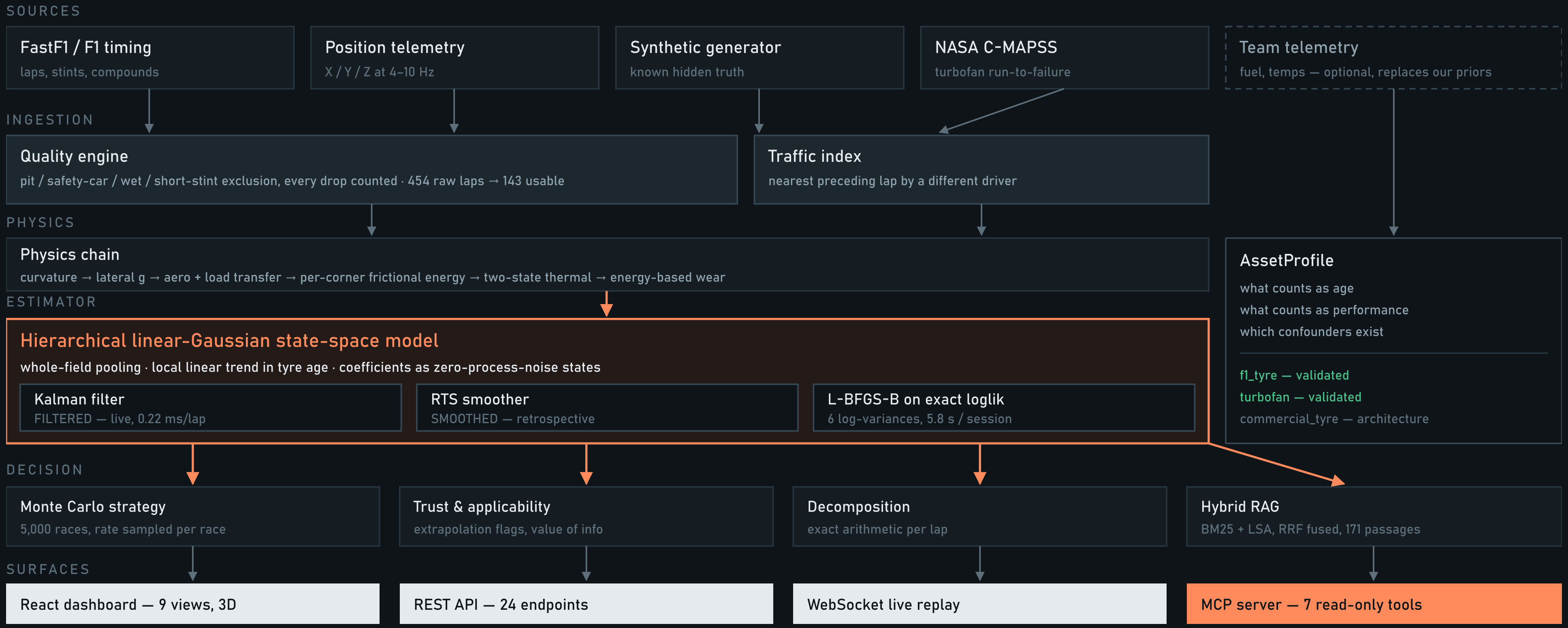
Full session fit

down from 25 s, identical results

Durbin & Koopman 2012 §6.4 · src/tyremind/models/ssm/kalman.py

SYSTEM ARCHITECTURE

One estimator, two modes, four consumers



The single warm block is the estimator. Two modes leave it — filtered for live, smoothed for retrospective — and they are never conflated. The `AssetProfile` on the right is what makes the estimator asset-agnostic: pointing it at turbofan engines required no change to the model itself.

17,278 lines Python · 6,583 lines TypeScript · 185 tests · runs fully offline

EVIDENCE 1 OF 5 · THE HEADLINE

Can it recover a rate it was never shown?

Twenty-five synthetic sessions with a hidden degradation rate per compound, buried under realistic fuel burn, track evolution, traffic, driver noise, scrubbed sets and a cliff. Neither estimator is told the answer. This is the only way to score *attribution* rather than prediction.

0.0044
TyreMind error
s/lap MAE · bias +0.0012

0.0966
Naive error
s/lap MAE · bias −0.0966

95.5%
Error reduction
75 comparisons

100%
Interval coverage
nominal 95% — conservative, and we say so

COMPOUND	TRUE RATE	TYREMIND	NAIVE	NAIVE BIAS	REDUCTION
HARD	0.041	**0.0035**	0.0983	−0.0983	96.4%
MEDIUM	0.072	**0.0039**	0.0980	−0.0980	96.0%
SOFT	0.115	**0.0057**	0.0934	−0.0934	93.9%

The most informative cell is the naive bias −0.0966 s/lap — almost exactly the fuel slope of 0.081, in the direction theory predicts. That is not a coincidence. It is the first collinearity showing up as a measured quantity. The naive method is not noisy; it is **systematically wrong by the size of the confounder it ignores**.

experiments/results/exp01_ground_truth_recovery.json · 25 seeds, 100% convergence

Does a Friday curve predict Sunday?

Degradation estimated from each event's FP2 and scored against that event's race. **No race data reaches the practice fit.** Practice and race differ in fuel load, traffic density, track state and driving style at once — genuine out-of-distribution transfer, not a holdout split.

0.0871
TyreMind error
s/lap MAE, 27 events, 2024+2023

0.1442
Naive error
s/lap MAE, same comparisons

+0.042
Systematic bias
reported, not tuned away

95%
Interval coverage
conformally calibrated, 62 comparisons

This number got worse, and that is the point

On five events this slide read **0.0518 s/lap**. On 27 events it reads 0.0871. The small sample was flattering us. What held is the comparison that matters — the naive method worsened in step, 0.1166 → 0.1442, and the **40% gap between them is stable**. A claim that survives a six-fold increase in evidence is worth more than a prettier one resting on ten comparisons.

The earlier version of this slide warned that a correction fitted on five events "would not survive the sixth". It did not. We tested nine explanations for the +0.042 s/lap bias with a multiplicity correction, found two that survive, built the causal story they implied — and then **refuted it**: practice runs are 4.4 laps *shallower* than race stints, not deeper, and forcing a common window made the error worse.

So the bias is corrected as a **measured offset, not an explained one**, and the interval is widened by conformal calibration until it covers what it claims. Three events are excluded because they were never dry-degradation sessions: Canada 2024 ran 67% of its race laps on wet rubber.

experiments/results/exp03_practice_to_race.json

Six models, two metrics — and they disagree

Lap-time prediction — the metric the field reports

MODEL	CRPS	COVER	BIAS DRIFT
Pooled regression	**0.525**	84%	+0.232
LightGBM	0.536	62%	−0.063
TyreMind	0.757	81%	**−0.483**
Fuel-corrected	0.985	76%	+0.443
Naive	0.991	76%	+0.497
Neural network (MLP)	2.034	69%	−3.582

Degradation recovery — the metric the product exists for

MODEL	RATE MAE	COVERAGE
TyreMind	**0.0041**	100%
Pooled regression	0.0068	78%
Fuel-corrected	0.0230	44%
Naive	0.0748	0%

LightGBM and the MLP cannot appear here. Neither has a parameter meaning "degradation rate" — nothing to score, nothing to report to an engineer, nothing to carry from Friday to Sunday.

The disagreement is the finding — and we ship it inside the product

Two rungs beat us on lap-time prediction — and on twenty races the leader is plain pooled regression, not LightGBM, which led when this table held four. **Neither of them has a parameter meaning "degradation rate" at all**, so neither can appear in the table beside it. LightGBM stays badly overconfident (62% coverage on nominal 95%). On extrapolation, bias drift is +0.232 for the lap-time leader against **−0.483** for the state-space model, whose error shrinks most as it forecasts past its training window. We previously called that the *only* such rung; at twenty races LightGBM is also roughly flat at −0.063, and we correct it here rather than restate it. The MLP is worst of all at −3.582, and it was tuned first across five configurations, because beating a badly-configured competitor would prove nothing. **That column is the physics-informed argument demonstrated rather than asserted:** encoding fuel as physics lets a model extrapolate it; learning fuel as a pattern does not.

experiments/results/exp05_model_ladder.json · expanding-window chronological folds

EVIDENCE 4 AND 5 · A FAILURE, AND A TRANSFER

One hypothesis that failed, and one that held on a second asset class

EXPERIMENT 04 — NEGATIVE RESULT, PUBLISHED

The energy clock did not work, and we know why

Predicted: replacing "tyre age in laps" with "cumulative tyre energy" as the degradation clock would improve transfer, because a tyre responds to energy through the contact patch rather than to distance.

Result: no meaningful difference. Mean R^2 gain −0.0008 across four stints.

Why, and this is the useful part: per-lap energy varies by only 2.3% within a stint. F1 drivers are extremely consistent, so cumulative energy is very nearly proportional to lap count and *no improvement is possible* within a stint.

This slide used to end by proposing the cross-session version — whether an energy clock absorbs the practice-to-race bias — as the next experiment. **It has since been run.** Nine candidate mechanisms, Benjamini–Hochberg corrected across all nine: the fuel-load reading was **not** among the two that survived, and the causal story built on those two was then refuted as well. Four hypotheses tested, four negative, each with a mechanism. That is more useful than a quiet retreat.

EXPERIMENT 07 — CROSS-DOMAIN VALIDATION

The same estimator on jet engines

F1 cannot prove this works: public data has no measured wear, so there is no answer to check against. **NASA's C-MAPSS turbofan benchmark does** — engines run to failure with published remaining-life labels.

The identical estimator, **no tyre-specific code**. "Laps" became flight cycles; "lap time" became a health index fused from 12 of 21 sensors; fuel and traffic switched off, because engines have neither.

22.7	40/100	32%
RUL RMSE, cycles	Held-out engines scored	Predicted early — the safe direction

We do not claim to be competitive. Purpose-built deep prognostics models reach 12–20 cycles. This is a tyre model pointed at engines with no retuning — a generalisation check, not a leaderboard entry.

It is now like-for-like. We score **all 100** engines in the FD001 test set, which is what published figures are quoted over. An earlier version scored 40, because the estimator fits the test set jointly and a 100-engine run would not converge; fitting in batches removes the limit. Re-running the original 40 in one batch reproduces 26.5 exactly, so the improvement comes from scoring the other 60 rather than from how the fit was split.

A check that is not circular: recovering circuit geometry it was never given

The chain runs GPS → path curvature → lateral acceleration → aerodynamic load and load transfer → per-corner frictional energy. Every step needed a validation that did not simply confirm itself.

The test: a clockwise circuit is mostly right-hand corners, and cornering right throws load onto the *left* tyres. So a clockwise circuit must show a left-side energy share above 50%. Rotation direction is published fact and is **never given to any code under test**.

CIRCUIT	PUBLISHED	LEFT-TURN ENERGY	CORRECT
Monza	clockwise	21.5%	yes
Barcelona	clockwise	25.9%	yes
Zandvoort	clockwise	29.5%	yes
Silverstone	clockwise	35.4%	yes
Austin	anti-clockwise	46.2%	no
Imola	anti-clockwise	60.1%	yes
Singapore	anti-clockwise	60.9%	yes
Interlagos	anti-clockwise	74.6%	yes

7 of 8, and the miss is reported as a miss

Austin at 46.2% is genuinely borderline — COTA's esses alternate direction, so the circuit really is close to balanced. Recovering this at all requires curvature sign, load transfer and energy integration to be correct **simultaneously**.

Where the textbook answer is wrong

Wear rises on **both sides** of the temperature window: too cold and the driver slides a stiff tyre and tears it (graining); too hot and the rubber abrades and blisters.

An Arrhenius form is the standard choice and cannot represent this, because it is monotone in temperature — it misses half of real tyre behaviour. We use a symmetric penalty around the working window, and guard it with a test so it cannot be silently "simplified" back.

What we refuse to claim

Slip angle, slip ratio, contact pressure, temperature and tread depth are **not observable from public telemetry at any price**. Frictional power is a proxy and every function returning one says `_proxy` in its name. Absolute watts are meaningless and are never reported.

experiments/results/exp06_circuit_asymmetry.json · docs/research/04_PHYSICS_FOUNDATION.md

THE PRODUCT

Nine views, built so a newcomer understands what is happening

Start here
Why the obvious method fails, on this session's real numbers, next to the fitted curve per compound with its uncertainty band.

Why is the car slow
Confounders lifted off a stint one at a time, animated. The tyre curve underneath is what is left standing.

Where it wears
The real racing line in 3D, coloured by tyre loading, with the friction envelope and per-lap load trace.

Tyre twin
Per-corner condition, competitive life remaining, and what-if counterfactuals: in clean air, on a fresh set.

When to pit
Five thousand simulated races per option, with the degradation rate **resampled from its posterior for every race**, plus the full pit-lap sweep.

Live monitor
Lap-by-lap streaming with no access to future data. Watch the interval collapse from ±0.200 to ±0.025 s/lap over 23 laps.

Does it work
The identifiability argument, the validation experiments and the practice-to-race scatter, in the product rather than in a paper.

Ask the method
Hybrid retrieval over the project's own research corpus — 171 passages, 18 files. Retrieves, deliberately does not generate.

Beyond racing
The turbofan result, engine by engine, and the three asset profiles — with the unvalidated one labelled as unvalidated.

Dark and light themes, fixed navigation, 3D racing lines, and every estimate drawn as an interval rather than printed as a bare number with a plus-or-minus appended. Runs entirely offline — a demonstration does not depend on conference wifi.

`apps/web · 6,583 lines TypeScript · React, ECharts, react-three-fiber`

Why an MCP server and a retrieval index, and not a chatbot

MCP — THE MODEL AS A TOOL AN AGENT CAN CALL

A dashboard competes for attention. A tool does not.

Every team already has analysts, and every analyst already has an AI assistant. If TyreMind is a website it competes for a race engineer's attention during a session. If it is a tool their assistant can call, it becomes part of a workflow they already have.

Seven tools, deliberately: `list_sessions`, `get_degradation`, `explain_lap`, `project_tyre_life`, `recommend_strategy`, `assess_trust`, `search_documentation`. The MCP practice literature is blunt that a server exposing forty tools degrades an agent before it has done anything.

Every tool returns uncertainty. An agent handed `0.113` will state it as fact; one handed `0.113 ± 0.023, and the model is extrapolating` can hedge correctly.

The server carries instructions that constrain the claim, telling the client in its own context that this estimates a latent performance state and not tread depth — because otherwise a helpful assistant will paraphrase "degradation" into "tyre wear" and overclaim on our behalf.

Every tool is read-only. No write path. The worst outcome of a confused agent is a confused answer, never a corrupted fit.

RAG — GROUNDING THE ANSWER TO "WHY SHOULD I BELIEVE THIS?"

Retrieval, deliberately without the generation

The honest answer to that question is spread across a research audit, a model card, a limitations register and seven experiment result files. Nobody reads all of it, so the question goes unanswered — or gets answered from memory, which is where confident wrong numbers come from.

A hybrid index over those files returns the passage verbatim with its source. **BM25 for exact terms** like "CRPS" or a specific number; **latent-semantic retrieval for paraphrase**, finding a passage about calibration when you ask "how sure is it". The two rankings are fused by **Reciprocal Rank Fusion** — by *position*, because BM25 relevance and cosine similarity share no scale and adding them would mean nothing.

The retrieval half is the half that carries the trust. It finds the paragraph; you read the paragraph. Add generation and the answer gets smoother, the citation becomes a gesture, and a wrong claim becomes indistinguishable from a right one.

A language model is used in exactly one place in this product: rewriting an already-computed explanation into plainer English, with every number fixed before it is called.

src/tyremind/mcp_server.py · src/tyremind/rag/index.py · docs/INTEGRATIONS.md

UNIQUENESS, STATED SO IT SURVIVES SCRUTINY

What is ours, what is standard, and what is explicitly someone else's

CAPABILITY	EXISTS ELSEWHERE?	THE CLAIM
Testing whether the attribution is correct	**No**	"We test whether the attribution is right, not just whether the prediction is close." Not: that we validated attribution on real F1 data — public data has no measured wear, so nobody can.
The track-evolution collinearity	**Not found**	"We characterise a second collinearity and resolve it with a physically motivated basis." Not: that we discovered something the field missed — it is a consequence of adding a term others did not model.
Pit stagger as identification	Partially	"Pit stagger across the field is a natural experiment separating tyre age from session lap. A single-driver model cannot use it." Multilevel F1 modelling exists; this application, we did not find published.
Per-corner energy from public telemetry	Not from public data	"Public position telemetry supports a loading estimate good enough to recover circuit geometry it was never given." Not: that we measure tyre loads.
Real-time online estimation	Yes — Pitwall runs live	"The same estimator serves both modes because it is recursive. An MCMC formulation could not." Architectural, not the filter.
Cross-domain transfer with ground truth	C-MAPSS is heavily studied	"The identical estimator predicts turbofan remaining life at 22.7 cycles RMSE, on all 100 test engines." Using it to validate a *motorsport* model is the unusual part.
Race strategy simulation	**Yes, thoroughly**	Not ours — Heilmeier et al. Only the input differs: the rate enters as a distribution, resampled per race.
State-space tyre degradation · MC pit optimisation · LLM race narration · Kalman, Archard, Pacejka, load transfer	**Published / textbook**	Not ours. Stated plainly, so that what *is* ours is credible.

> Existing work predicts lap times well and never checks whether the reason it gives is the right one. **We characterise why that check is hard** — three collinearities, two resolvable only by assumption — build an estimator that handles them, and test it against a truth we control, on a second asset class with real ground truth, and on the practice-to-race transfer the challenge actually asks for.

LIMITATIONS — PLACED BEFORE THE COMMERCIAL CASE, DELIBERATELY

Where TyreMind is wrong, or would be

Situations where it should not be trusted

SITUATION	WHAT THE PRODUCT DOES ABOUT IT
Wet or drying track	Different physical process; priors do not describe it. Wet compounds excluded at ingestion; residual noise visible in diagnostics.
Past the observed tyre age	Applicability score decays; below 50% the interface says the model is extrapolating.
A briefly-run compound	Estimate dominated by its prior. Lap count shown, `assess_trust` flags it.
Safety car, red flag, VSC	Removed by a robust MAD threshold — and counted, never silently dropped.
Single-car analysis	Loses the stagger identification. Measured: halving the field moves error 0.0047 → 0.0073.
Puncture, debris, damage	Not in the model. Innovation z-scores surfaced live — sustained same-signed innovations mean the model is being surprised.

Calibration is imperfect — and in both directions

100% coverage against a nominal 95% on the degradation task is **not a perfect score**: intervals are conservative, wider than strictly necessary.

On lap-time prediction it goes the other way — our 95% intervals cover only **73%**, and pooled regression does better at 79%. Every rung in that ladder undercovers, but ours is not the best of them. A reader is entitled to both numbers, not only the flattering one.

Deliberately not built

Full Pacejka identification — slip angle and slip ratio are not observable, so the parameters would be unconstrained by any available data.

MCMC — no incremental mode, which would have killed the real-time claim.

A validated fleet product — no public dataset pairs tread depth with telematics. Building that claim without one would be dishonest.

The honest summary

It does not measure tyre wear. It cannot be validated against real tyre wear, because no such public data exists. It should never be used for a safety decision. And on the narrow task of predicting lap times, a gradient-boosted tree beats it — while being unable to answer the question the product exists to answer.

Nine bugs that produced plausible wrong answers

A model that fails loudly is easy. The danger is one that fails *quietly and confidently* — every entry below did, and each was caught only by a test or a cross-check.

BUG	SYMPTOM	FOUND BY
Falsy-zero on session lap	Track basis pinned at zero, so all track evolution leaked into degradation as a uniform −0.025 s/lap offset. **Invisible on real F1 data**, which numbers laps from 1.	Live-vs-batch agreement test
Traffic index identically zero	Grouped by lap number, but two drivers' "lap 5" can be twenty minutes apart. The confounder was in the model and contributing nothing.	Inspecting the fitted coefficient
Session progress recomputed per frame	Test blocks measured progress from their own start. No error, no leak — just wrong by tens of seconds, making two benchmark rungs look worse than they are.	An implausible 33 s MAE
Unregularised pooled regression	One Silverstone fold at 176 s MAE while every other fold was under a second.	Per-fold inspection
include_router adding nothing	Eight API endpoints registered zero routes, with no error raised. Every one would have 404'd in the demo.	Listing routes after registration
FastF1 resolving Interlagos to Zandvoort	Analysis of an entirely different circuit, with a warning rather than an error.	Two circuits producing byte-identical results
Inverted confidence interval	`competitive_life_interval()` returned (8, 1) — a lower bound above its upper bound, shown to the reader as a range.	Reading the output
Frozen chart clip path	An interrupted animation left a line's clip frozen part-way, so a 43-lap stint rendered as 14.	Reading pixels back from the canvas
Stale experiment result	A committed result file predated the RUL cap added to its own script, overstating cross-domain error at 56 cycles instead of 26.5.	Cross-checking a chart against its script

Second route to the same number

The recursive likelihood is checked against an independently constructed joint multivariate normal, to 1e−9 — a different implementation, not a regression snapshot.

Non-circular physical checks

Circuit rotation is published fact never given to the code. Recovering it needs curvature sign, load transfer and energy integration all right at once.

Truth we control

The synthetic generator is the oracle — the only way to score attribution.

`185 tests · ruff clean · every number read from a committed result file`

IMPACT AND ADOPTION

Who needs this first, and why integration is nearly free

SEGMENT	WHY THEY NEED IT, AND WHAT THEY GET ON DAY ONE
Junior formulae F2, F3, F4, regional	Spec series, no tyre telemetry, small engineering staffs. The **identification problem binds hardest exactly here.** A degradation estimate with honest intervals from timing data alone. Highest need, lowest data, no incumbent tool — this is the beachhead.
Customer & privateer teams GT3, endurance, TCR	Long stints, driver changes, tyre life as the dominant strategic variable. Stint-length planning under uncertainty, plus per-corner energy explaining why a compound behaves differently between circuits.
F1 customer teams & analysts	Telemetry but not always a dedicated tyre modelling group. A structurally different second opinion — valuable precisely because it sometimes disagrees — and the ability to replace our fuel prior with their measured fuel and watch the interval narrow.
Broadcast & fan analytics	"The tyres are going off" is asserted every Sunday with no number behind it. A live per-car figure with an interval is the first version of that claim that can be checked. Lowest integration cost of any segment.
Tyre manufacturers	Cross-circuit degradation comparison with the circuit-loading confound made explicit rather than left in the residual.

What a wrong rate costs

A stop costs ~20 s of track position. On a representative Monza race state the product reports a **seven-lap window in which any stop costs under one second** against the optimum — and a stay-out option beating every stop by 9.7 s.

That second number is the more useful one. The value is not only picking the right lap; it is knowing *how much the choice matters*, so a strategist spends attention where the curve is steep.

1 · No new data required

The baseline model consumes lap times, stints and compounds — what every timing system already produces. Telemetry improves the physics layer but is not required for the degradation estimate.

2 · No new screen required

MCP makes the estimator a tool the team's existing assistant can call.

3 · Priors are replaceable, not baked in

Substitute measured fuel for our prior and the largest assumption in the method disappears. The architecture was designed for that substitution from the start.

Strip the motorsport words away and it is not a tyre problem

Something wears out while it works, you cannot see its condition directly, and the signal you can see is contaminated by conditions that also change over time.

That fits a jet engine, a truck tyre, a battery, a bearing, a cutting tool and a wind-turbine gearbox. So the estimator consumes an `AssetProfile` — what counts as age, what counts as performance, which confounders exist — rather than a tyre. Pointing it at turbofans required **no change to the model**, which turns "it would generalise" from a slide into a property of the code.

DOMAIN	AGE IS	PERFORMANCE IS	THE CONFOUNDERS	STATUS
Formula 1 tyre	Laps	Seconds per lap	Fuel, track evolution, traffic, driver	**Validated on recovery against known truth**
Turbofan engine	Flight cycles	Health index, 12 sensors	Operating mode	**Validated — 22.7 cycles RUL RMSE**
Commercial vehicle tyre	Thousand km	Rolling resistance index	Payload, weather, route, driver	Architecture only — no public dataset pairs tread depth with telematics
EV traction battery	Equivalent full cycles	Usable capacity	Temperature, charge rate, depth of discharge	Same shape — confounders stronger than the signal, which is the case this method is for
Wind turbine gearbox Machine tool insert	Revolutions Cut metres	Vibration signature Spindle load	Wind regime, curtailment Material batch, feed and speed	Same shape — site-level pooling, or part-to-part stagger, replaces pit stagger

What we refuse to claim. There is no validated fleet result, because no public dataset pairs tyre tread depth with vehicle telematics — we looked; it does not exist. The product shows fleet arithmetic labelled as arithmetic, **on the screen rather than in a footnote**. Rows below the two validated ones are architectural claims about problem shape, not measured results.

`src/tyremind/assets/profile.py · exp07_cross_domain.json`

What already scales, and what has to change

Already scales — measured, not assumed

PROPERTY	MEASURED	WHY IT HOLDS
Per-lap live update	0.22 / 0.74 ms	A recursive filter carries a fixed-size state, so a lap in hour three costs what lap one cost.
Full session fit	5.8 s	Structured sparse transitions make propagation $O(n^2)$ not $O(n^3)$ in grid size.
Strategy simulation	50,000 races in 33 ms	10,000 per option across 5 options; the interface uses 5,000 per option — 25,000 races, 17 ms. Fully vectorised with common random numbers, so comparisons are noise-free at any count.
Offline operation	8 sessions cached	No network dependency at run time — the property that makes a demo safe makes trackside safe.

The scaling shape of the estimator

$O(n^2)$ per lap in tracked states — two per active car plus a small shared set. A 20-car grid is ~125 states, and 0.22 ms leaves four orders of magnitude of headroom against an 80-second lap. A hundred-car endurance field is ~25× the arithmetic and still comfortably real-time.

The binding constraint at scale is not compute. It is data quality and prior calibration per series.

Has to change, in order

	CHANGE	APPROACH
1	**Live timing feed adapter**	The estimator already consumes one lap at a time through a stable interface; only the source changes.
2	**Session-state persistence**	Filter state is a mean vector and a covariance — a few hundred KB. Checkpoint per lap; resume exactly, because the recursion is Markov in that state.
3	**Multi-tenant isolation**	Per-tenant store and fit cache. The estimator is stateless between fits, so isolation is storage, not modelling.
4	**Per-series priors**	Fuel and track priors are F1 values. Promote the physics config to a per-series profile, each shipping its own measured cost-if-wrong table.
5	**Warm-started hyperparameters**	Seed from the previous session at the same circuit — a circuit's variance structure is stable across a weekend.
6	**Model monitoring**	Innovation z-scores already exist. Add alerting on sustained same-signed innovations, plus per-event tracking of the practice-to-race bias.
7	**Season-wide backfill**	Five events is a small validation set. Run the practice-to-race experiment nightly over a full season and publish the drift.

measured on a 20-car grid, 921 laps

ROADMAP

From prototype to deployed system — with the research questions in the right order

NOW — prototype complete	0–3 months — one live series	3–9 months — multi-series	9–24 months — second industry
Estimator, physics, validation suite 9 views, 24 endpoints, 185 tests MCP server, hybrid RAG C-MAPSS transfer validated Runs fully offline	Live timing feed adapter Session-state checkpointing Per-series prior calibration Season-wide backfill of the practice-to-race claim Cross-session energy clock test	Multi-tenant isolation Warm-started hyperparameters Innovation-based drift alerting Prior replacement from team fuel telemetry Thermal state from IR where a series provides it	Second asset class with its own ground truth — battery or gearbox Per-domain sensitivity suite On-premise enterprise deployment Published identifiability analysis for degradation under confounding

The research questions, in the order they should be answered

1 — Does an energy clock absorb the +0.047 s/lap practice-to-race bias *across* sessions? Within a stint it provably cannot; across sessions it is untested.

2 — Does replacing the fuel prior with measured fuel mass narrow the interval as much as the sensitivity analysis predicts? The cheapest way to falsify our largest assumption.

3 — Does whole-field pooling still identify when the field is small? Halving the grid moved error 0.0047 → 0.0073; a ten-car junior series is the real test.

What would falsify our central claim

Stated because a roadmap that cannot be wrong is a wish list.

— A flexible learner recovering the hidden rate as accurately as we do. Today it cannot, because it has no such parameter — but a structured neural state-space model would have one.

— The practice-to-race bias failing to hold its sign across a full season, meaning it is not the fuel-load effect we believe it is.

— Interval coverage falling below nominal on a larger real sample, meaning our conservatism is an artefact of the synthetic generator.

IN ONE PARAGRAPH

The brief asked for a model that strips external noise from practice sessions to produce clean degradation curves, and validates them against race-day pace. Our argument is that the difficulty is not *building* such a model — it is **knowing whether you have**. Every number in this deck exists to answer that second question, including the ones that do not flatter us.

95.5%
Error reduction recovering a hidden rate
where the standard method has the wrong sign

0.0871
Practice→race error, s/lap
against naive 0.1442, on the brief's own question

0.22 ms
Per-lap live update
the same estimator, forward-only

22.7
RUL RMSE on turbofans
the same estimator, a different industry

TYREMIND

17,278 lines of Python · 6,583 of TypeScript · 185 tests · 24 endpoints
13 recorded experiments · 9 views · MCP server · hybrid RAG · runs fully offline

Every figure in this deck is read from a committed result file, not typed by hand.
Regenerating the experiments regenerates the claims. The product's own retrieval index searches the same files, so any number here can be traced to the document it came from — including the failed hypothesis, the systematic bias, and the competitor that beats us.